

Schematic → interactive 3D: what's actually possible (and what isn't)

The honest core: a 2D line drawing has no depth data. Accurate 3D must come from real 3D sources — never invented from a scan. Five approaches, grounded → speculative.

1 · Interactive 2D — deep zoom + clickable callouts

RECOMMENDED · GROUNDED

Tile-zoom the REAL scanned schematic (like a map); auto-place clickable hotspots on the FIG callout numbers (from OCR coordinates) that jump to the part row (NSN/part#). Pan, deep-zoom to fine detail on the authoritative drawing.

Accuracy = the manual itself · Effort ~2-4 wks · Needs schematic OCR done

Delivers the real job: inspect fine features + identify the exact part. Not rotation, but honest.

2 · Multi-view switcher (where the TM has them)

GROUNDED, LIMITED

Many TMs show front/side/section views. Stitch the views the manual already provides into a view switcher — several real angles, no invented geometry.

Accuracy = real drawing views · Effort ~1-2 wks

Pseudo-3D from real art · only the angles the manual provides.

3 · True 3D from REAL CAD / 3D-TM assets

ACCURATE 3D — if files exist

Render actual 3D models (glTF/STEP→glTF, IETM/S1000D or OEM CAD) in a rotatable viewer, linked to the parts index. Real rotation, zoom, pick-a-part.

Accuracy = the CAD · Viewer ~2-3 wks · BLOCKER: sourcing the files (controlled/availability)

True rotation & feature zoom · depends on obtaining 3D data — an org/supply problem, not code.

4 · True 3D via photogrammetry of the real part

ACCURATE 3D — capture workflow

Photograph the actual part (many angles) → reconstruct an accurate textured mesh (COLMAP/RealityCapture). Build a library over time, prioritising look-alike parts. Rotate any captured part, zoom to holes/ports/connectors.

Accuracy = the real part · Pipeline ~3-5 wks + per-part capture labor

The genuine 'rotate & zoom to the tiny difference' experience · grows part-by-part.

5 · AI single-image → 3D from the scan

NOT RECOMMENDED

Neural image-to-3D would generate a plausible mesh from one drawing — but it INVENTS geometry (wrong hole count, wrong port). For parts ID where the tiny difference is the whole point, a confident-but-wrong model is dangerous.

Accuracy = hallucinated · Fast to demo, wrong in practice

Breaks the project's never-invent rule. The exact failure mode you care about (holes/ports) is what it gets wrong.